Nuclear Data Uncertainty Quantification Using the VENUS-3 Benchmark Experiment

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Abstract. Understanding how nuclear data uncertainties propagate in reactor physics calculations is essential for improving the reliability of pressurized water reactor (PWR) analyses. This work studies the behavior and impact of nuclear data and their uncertainties in the VENUS-3 PWR pressure vessel benchmark using a Fast Total Monte Carlo (Fast-TMC) approach with MCNP6.3 and modern evaluated nuclear data libraries, including the most recent releases from ENDF/B (VIII.1) and JEFF (4.0). Results show good agreement with experimental reaction rates and relatively small nuclear data-induced uncertainties for most libraries. In contrast, JEFF-4.0 initially exhibited noticeably larger uncertainty estimates; further analysis identified the issue as originating from the ^{16}O covariance data processing with NJOY. Removing the affected isotope or reprocessing its covariance data with AMPX substantially reduced the uncertainties to levels consistent with other libraries. Overall, the small nuclear data uncertainties observed in VENUS-3 leave little room for data assimilation to improve PWR calculations and highlight the importance of validating covariance data processing for reliable uncertainty quantification.

1 Introduction

Nuclear data are the basis of neutronic simulations used to predict key quantities in reactor physics, such as neutron flux and fluence in reactor pressure vessels, which directly influence estimates of material embrittlement and structural integrity over reactor lifetime. Uncertainties in these data propagate through transport calculations and can ultimately affect safety margins and operational decisions in power reactors [1].

Benchmark experiments like VENUS-3 [2], designed to represent the neutron flux attenuation in PWR system and providing a detailed estimate of the experimental uncertainty, provide an opportunity to evaluate how nuclear data uncertainties propagate to integral responses, thereby guiding uncertainty quantification (UQ) in full-scale PWR analyses. By comparing calculated-to-experimental (C/E) reaction rate ratios and their associated uncertainty bands, such benchmarks allow assessing both the predictive capability of evaluated nuclear data libraries and the practical significance of their uncertainties in reactor-relevant conditions.

The present work studies the impact of evaluated nuclear data uncertainties on reaction rate predictions in the VENUS-3 benchmark and examines how recent updates in major evaluated libraries influence both the C/E agreement and the associated uncertainty estimates. The analysis compares the most recent ENDF/B-VIII.1 [3] and JEFF-4.0 [4] releases with their preceding versions, ENDF/B-VIII.0 [5] and JEFF-3.3 [6], using the Fast Total Monte Carlo (Fast-TMC) [7] approach to propagate cross-section covariances through Monte Carlo transport simulations.

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2 Methodology

The VENUS-3 experiment was a reactor assembly designed to reproduce the geometry and neutron field of a pressurized water reactor core with a steel pressure vessel wall. The core used uranium dioxide fuel enriched to 3.3% in ^{235}U and included four peripheral assemblies equipped with partial-length shielding (PLSA). In these assemblies, the upper half of the fuel rods were replaced by solid stainless steel, which significantly reduced the fast neutron flux reaching the vessel. This configuration enabled the study of neutron attenuation through structural materials and the validation of three-dimensional transport and dosimetry calculation tools. Reaction rates were measured at 386 positions using ^{58}Ni(n, p), ^{115}In(n, n'), and ^{27}Al(n, α) activation foils located both within the core and in ex-core regions near the vessel [2] (see Figure 1). A detailed description of the benchmark geometry, materials, and measured reaction rates is given in the NEA benchmark report [8]; the experiment is also distributed through the SINBAD shielding benchmark database [9]. The experimental uncertainty depends on the detector type. The benchmark report quotes about 6% for the ^{115}In(n, n') and ^{58}Ni(n, p) reactions and about 7% for ^{27}Al(n, α) [8]. These values are upper bounds, given for the detectors at the reactor barrel, and they decrease for positions closer to the core. For simplicity, the C/E comparison applies a single band per reaction, set to these upper-bound values.

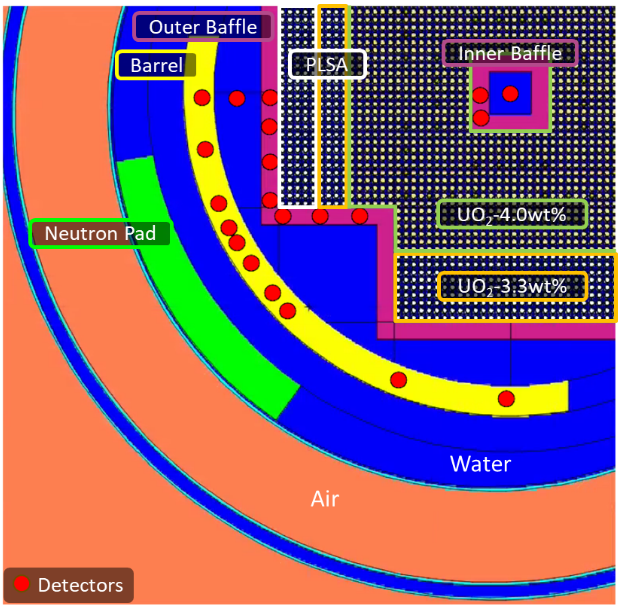

Figure 1. Radial cross-section of the VENUS-3 experimental configuration showing main components and detector locations. Only some detector positions are displayed for clarity.

The transport calculations were performed with MCNP6.3 [10] in criticality (KCODE) mode. A preliminary run was first carried out to obtain a converged fission source, using 5×10^5 particles per cycle over 500 cycles; the resulting SRCTP file was used as the starting source for all subsequent simulations. Each Fast TMC simulation then started from this converged source with 2×10^6 particles per cycle, discarding the first 20 cycles as inactive to let the source re-settle and accumulating the reaction-rate tallies over the following 200 active cycles. The multigroup covariance matrices used for sampling were processed with NJOY [11] ERRORR module in a 56-energy-group structure. Reaction rates were computed using cross sections from the IRDFF-II dosimetry library [12], ensuring consistency with experimentally measured activation foils. Propagation of uncertainties was carried out for all reactions and isotopes relevant to the attenuation path, excluding fuel and cladding isotopes. Only reactions with available covariance data in each library were perturbed: 30 isotopes in JEFF-3.3, 44 in JEFF 4.0, 26 in ENDF/B-VIII.0, and 29 in ENDF/B-VIII.1. The perturbations were performed using KIKA (`kika-nd`, v0.1.2) [13], an open-source Python package, which was used to generate perturbed ACE-format files. For each library, 1024 perturbed nuclear data sets were produced using Sobol low-discrepancy sequences within the Fast TMC framework. This approach enables the separation of nuclear data and statistical contributions to the total response variance. Reliable separation requires that Monte Carlo uncertainty remains below 50% of the total observed uncertainty [14]; otherwise, only an approximate value of the nuclear data uncertainty or an upper bound combining both sources can be obtained.

3 Results

The calculated-to-experimental (C/E) reaction rate ratios obtained with the four evaluated libraries are summarized in Table 1. The table presents the mean C/E, the root-mean-square error (RMSE), and the percentage of detectors whose calculated values lie within the one-sigma experimental uncertainty of 6–7%.

Table 1. Mean C/E values, root-mean-square error (RMSE), and percentage of detectors within 1σ experimental uncertainty for each evaluated library.

Library	Mean C/E	RMSE	Within 1σ (%)
ENDF/B-VIII.0	0.9662	0.0620	70.47
ENDF/B-VIII.1	0.9991	0.0428	93.26
JEFF-3.3	0.9906	0.0450	91.16
JEFF-4.0	0.9928	0.0456	92.75

Among all libraries, ENDF/B-VIII.1 provides the best overall agreement with experiment, with a mean C/E of 0.999 and more than 93% of detectors within experimental uncertainty. Both the ENDF/B and JEFF series improve over their previous releases. The largest gain, from ENDF/B-VIII.0 to ENDF/B-VIII.1, follows the revised ^{56}Fe cross sections that dominate neutron transport through the steel. ENDF/B-VIII.1 and JEFF-4.0 share this recent ^{56}Fe evaluation and both reach a mean C/E near unity, whereas ENDF/B-VIII.0, which keeps the earlier data, remains about 3% low; JEFF-3.3 also lies close to unity. That two independently maintained evaluations agree once they adopt the same ^{56}Fe data supports the attribution, which the spatial pattern examined below reinforces.

Grouping the detectors by reaction and by position helps interpret the C/E spread (Figure 2). On average, the ^{115}In(n, n') detectors read about 2–3% below unity in the three recent libraries, and lower still in ENDF/B-VIII.0, whose whole set is shifted down. The ^{58}Ni(n, p) detectors located in the 3.3% enriched fuel inside the core (detectors 189–244 in Figures 4 and 5) read about 3% high in every library; this offset is nearly identical across the four libraries, which suggests it originates in the dosimetry cross sections or the experimental normalization. The ^{27}Al(n, α) reaction shows by far the largest scatter, mainly due to the higher statistical uncertainty of its tally: its high energy threshold scores few reactions and inflates the variance. The C/E change between ENDF/B-VIII.0 and ENDF/B-VIII.1 follows the amount of steel along the neutron path: it is smallest for the in-core fuel detectors (about 0.4%) and grows to about 5% for the outermost detectors at the core barrel (Figure 3). The detectors behind the partial-length shielding assemblies show only an intermediate change. This is consistent with the revised ^{56}Fe cross sections being the main driver, although the two releases differ in more than ^{56}Fe, so a dedicated single-isotope calculation would be needed to confirm it. No clear axial or azimuthal dependence is found. Across the ^{58}Ni(n, p) detectors along the steel path (roughly detectors 1–100), JEFF-4.0 shows more detector-to-detector structure in the

C/E than ENDF/B-VIII.1; identifying the isotope responsible would require a dedicated isotope-by-isotope study.

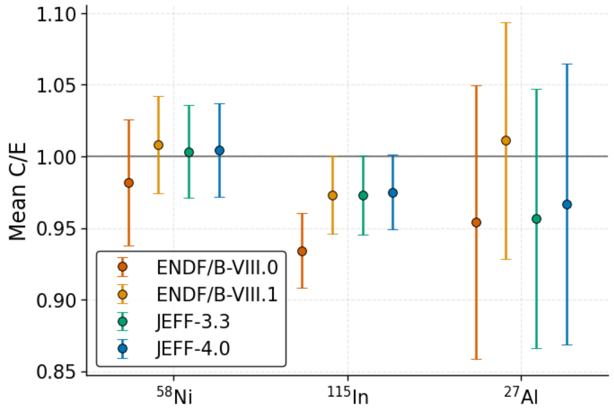

Figure 2. Mean C/E by reaction for the four libraries. Whiskers are the standard deviation across detectors.

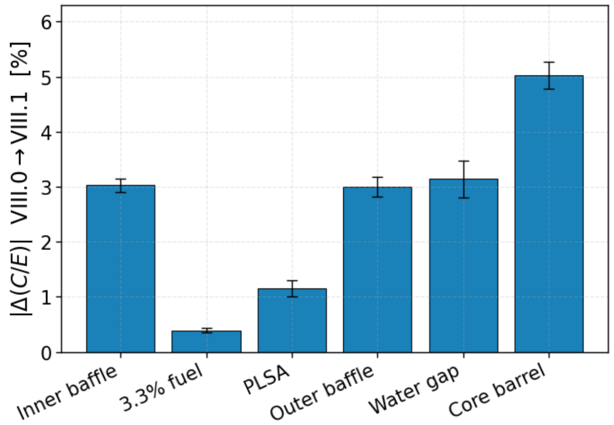

Figure 3. Change in the ^{58}Ni C/E between ENDF/B-VIII.0 and ENDF/B-VIII.1, by detector region ordered from the core to the vessel. The change grows with the steel thickness along the neutron path. Error bars are the standard error of the mean.

Uncertainty quantification results obtained through the sampling approach are presented in Figures 4 and 5 for ENDF/B-VIII.1 and JEFF-4.0, respectively. The plots show the one-sigma nuclear data uncertainty obtained from the spread of the sampled results. Fast TMC can isolate this nuclear data contribution from the Monte Carlo statistical noise of each run only when the statistical uncertainty stays below 50% of the total observed spread [14]. Pink points mark the detectors where this limit is not met: there the nuclear data uncertainty, σ_{ND}, is smaller than the residual statistical noise, σ_{Stat}, and cannot be separated reliably, so the plotted value is the observed total one-sigma, σ_{Tot}, an upper bound that contains both the nuclear data and statistical contributions. These pink-marked detectors already carry low statistical uncertainty, and the nuclear data part is smaller still; separating it would require much longer runs to further reduce the statistical noise, at a high computational cost and with little practical value, since the nuclear data uncertainty is already small. For ENDF/B-VIII.0, ENDF/B-VIII.1, and JEFF-3.3, the propagated nuclear data uncertainties are consistently low, with average values between 2% and 4%. When including the statistical contribution, the overall mean uncertainty remains below 7%. In contrast, JEFF-4.0 shows significantly larger propagated nuclear data uncertainties, with an average around 10%.

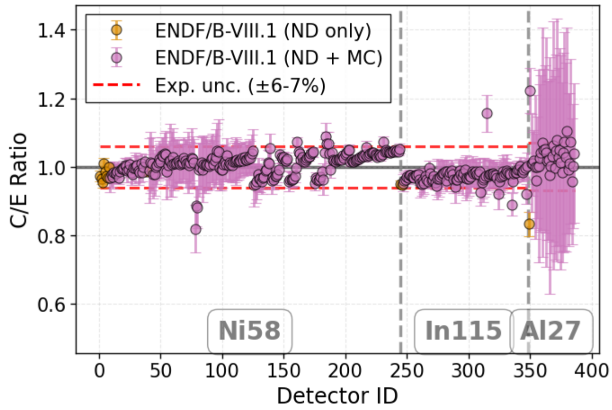

Figure 4. Calculated-to-experimental (C/E) ratios and one-sigma propagated uncertainties in orange, σ_{ND}, and in pink σ_{Tot},for ENDF/B-VIII.1.

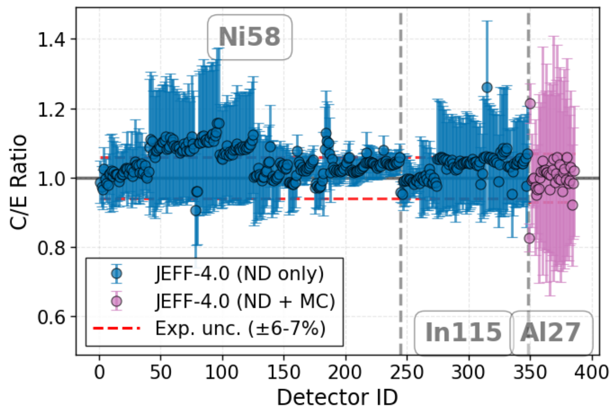

Figure 5. Calculated-to-experimental (C/E) ratios and one-sigma propagated uncertainties for JEFF-4.0 in blue, σ_{ND}, and in pink σ_{Tot}.

The source of this increased uncertainty was traced to the ^{16}O covariance data in JEFF-4.0. During NJOY processing, the multigroup covariance matrices for ^{16}O displayed unrealistically large variances in the elastic scattering and radiative capture reactions, exceeding 1000% for elastic scattering at energies above 1 MeV. This behavior was consistent with the higher uncertainties observed in detectors located in regions where water plays a dominant role in neutron transport, further supporting the hypothesis. Additional simulations were performed to verify this effect, both by removing ^{16}O from the sampling and by reprocessing its covariance data with the PUFF module of

AMPX [15]. In both cases, the resulting propagated uncertainties aligned closely with those obtained for the other libraries, particularly JEFF-3.3.

4 Conclusions

This work quantified the impact of nuclear data uncertainties on the VENUS-3 benchmark and assessed the performance of recent nuclear data library updates for predicting reaction rates relevant to PWR pressure vessel fluence calculations. The results show good agreement between calculations and experimental data, with mean C/E ratios close to unity for all modern libraries and visible improvement between consecutive releases. The marked gain from ENDF/B-VIII.0 to ENDF/B-VIII.1 is consistent with the updated ^{56}Fe cross sections, since the change between the two releases grows with the amount of steel along the neutron path. Some detector-to-detector C/E trends seen with JEFF-4.0 are not explained by ^{56}Fe and warrant a dedicated isotope-by-isotope study. Uncertainty propagation using the Fast-TMC approach demonstrated that nuclear data uncertainties in VENUS-3 are small, typically between 2% and 4% for most libraries. These values fall below the experimental uncertainties of 6–7%, so the benchmark offers limited sensitivity for reducing nuclear data covariances through data assimilation in PWR ageing studies. Because its core, steel, and water geometry is closer to a power reactor than single-material shielding benchmarks, VENUS-3 provides a first realistic estimate of the nuclear data uncertainties to expect on PWR pressure vessel fluence. Within these uncertainties, the calculations remain consistent with the VENUS-3 measurements for both the nominal data and their covariances.

The initially high propagated uncertainties observed for JEFF-4.0 were traced to an issue in the processing of ^{16}O covariance data. After reprocessing the affected matrices with AMPX, the resulting uncertainties became consistent with those from JEFF-3.3. This similarity is coherent with the fact that covariance data in the two JEFF evaluations are largely unchanged. Future work could extend this analysis to additional benchmarks such as H.B. Robinson, ASPIS, and PETALE. This study propagated cross-section covariances; angular-distribution covariances were not included. Further efforts will focus on propagating uncertainties in angular distributions, which may have a significant impact on attenuated neutron problems relevant to reactor pressure vessel environments.

Acknowledgements

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